Nihar Majalikar

CCNA-certified networking | infrastructure | security | automation

LinkedIn: linkedin.com/in/nihar-ravi-majalikar-1bb75b199/

GitHub: github.com/NiharMajalikar Portfolio: niharmajalikar.github.io/homelab-portfolio/

PROFESSIONAL PROFILE

CCNA-certified networking and infrastructure practitioner with a Master of Engineering in Telecommunications and Networking. Evidence-backed work spans routing and switching, GIS-assisted industrial communications research, segmented operational technology architecture, Linux infrastructure, virtualisation, monitoring, backup automation and CI/CD.

CORE TECHNICAL SKILLS

Networking: TCP/IP, routing, switching, VLANs, static routing, RIP, OSPF, BGP fundamentals, DNS, WAN concepts, packet analysis and troubleshooting

Security and access: SSH, ACLs, IPsec, WireGuard/Tailscale, firewall configuration, TLS, certificates, PKI concepts and OT segmentation

Infrastructure and automation: Linux, Proxmox, KVM/QEMU, Docker, Docker Compose, Pi-hole, Uptime Kuma, Nextcloud, MariaDB, Bash, cron and GitHub Actions

Microsoft cloud management: Microsoft 365 endpoint security, Intune device management, Microsoft Entra ID identity and access management, and Azure identity concepts

Analysis and documentation: QGIS, ns-3, FlowMonitor, NetAnim, CelPlanner, Cisco Packet Tracer, Python, C++, pandas, matplotlib and technical reporting

CERTIFICATION

Cisco Certified Network Associate (CCNA 200-301)

SELECTED TECHNICAL PROJECTS

GIS-assisted hybrid communications and segmented OT architecture

Master's thesis project | QGIS | ns-3 | Cisco Packet Tracer | C++ | Python

- Designed and evaluated a GIS-assisted hybrid communication architecture for a lagoon-based wastewater treatment plant monitoring environment.
- Compared LPWAN, Wi-Fi HaLow, 5G or fibre, and fibre backhaul options for connecting sensors, CCTV nodes, RTUs, gateways and a central server.
- Modelled 423 infrastructure nodes and 395 communication links, then evaluated packet delivery, delay, throughput and loss in simulation.
- Validated VLAN separation, redundant core paths, gateway routing, a proposed OT DMZ and restricted firewall flows in Packet Tracer.

Personal infrastructure, monitoring and CI/CD homelab

Proxmox | Ubuntu Server | Docker Compose | Pi-hole | Uptime Kuma | Nextcloud | Tailscale

- Built a Proxmox-hosted Ubuntu Docker environment for DNS monitoring, availability checks, private cloud storage and container management.
- Implemented private remote access, weekly backups of application data and configuration, plus a monitored backup-health signal.
- Documented a GitHub Actions deployment path using Tailscale, SSH and Docker Compose for a containerised homelab status application.

SELECTED TECHNICAL PROJECTS - CONTINUED

Intermediate Network Engineering laboratory sequence

CSE5INE(BU-1) | 11 supervised university practical labs | Cisco IOS | Packet Tracer

- Progressed from VLSM and basic router configuration through static routing, RIPv1/v2, single- and multi-area OSPF, DR/BDR behaviour, eBGP and a multi-protocol capstone.
- Used interface state, routing tables, neighbour status, ping and traceroute to verify paths and troubleshoot configuration or advertisement faults.

LTE-Advanced radio network planning and optimisation

ELE5001 | CelPlanner | University simulation and planning project

- Planned an LTE-Advanced radio access network across an 18.5 x 15.5 km assigned service area using 18 tri-sector eNodeB sites and 54 cells.
- Evaluated coverage, interference, data rate, traffic demand and mobility; the simulation reported more than 91% downlink area above -90 dBm and C/I above 20 dB across 90% of the service area.
- Identified high session rejection under heavy simulated demand and proposed capacity-focused optimisation steps without claiming a live deployment.

Network Systems and Web Security laboratory portfolio

CSE5NSW | 10 documented controlled laboratory and simulation exercises

- Configured and analysed DNS, HTTP/HTTPS, FTP, email, NTP and AAA services in Cisco Packet Tracer.
- Compared Telnet and SSH traffic, implemented standard and extended ACLs, and analysed a remote-access IPsec VPN path.
- Explored hashing, signing, certificate verification and PGP/GPG encryption, then applied access controls to simulated cloud and IoT scenarios.

EDUCATION

Master of Engineering - Telecommunications and Networking

La Trobe University, Bundoora VIC | Jul 2024 - Jul 2026

- Project work includes telecommunications planning, network simulation, segmented operational technology architecture and network security.

Bachelor of Engineering - Electronics and Communication Engineering

N.M.A.M. Institute of Technology, Nitte, India | Graduated Jun 2023

EXPERIENCE

Management and Site Intern

Goa Shipyard Ltd, Vasco da Gama, India | Jul 2022 - Aug 2022

- Supported equipment inspections, calibration checks and field observations, then documented technical issues and maintenance findings.
- Assisted engineering staff with electronic and electrical equipment checks, including pressure gauges and EM-log systems.

Container Unloader - Casual on-hire

HireGalaxy Personnel | Host: Viewscape, Bayswater VIC | Nov 2025 - Jun 2026

- Worked safely in a fast-paced warehouse environment while following site induction, WHS, PPE and manual-handling requirements.

Privacy note: This public edition intentionally omits a street address, postcode, personal email address and phone number. Please use LinkedIn for contact. Project evidence is available through the portfolio link on page 1.